

ABHI JAIN

Boston, MA

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EDUCATION

Boston University

September 2022 - May 2027 (expected)

Ph.D. Biostatistics

- Dissertation: Bayesian tree-based methods for causal inference with spatial data
 - Advised by Dr. Shariq Mohammed and Dr. Sara Lodi
- Relevant coursework: Estimation Theory, Hypothesis Testing, Linear Models, Correlated Data Analysis, Nonparametric and Semiparametric Data Modeling, Statistical Analysis of Point Process Data, Causal Inference, Machine Learning

Wake Forest University

August 2020 - May 2022

M.S. Statistics

- Master's Thesis: Using Zero-inflated Models to Estimate the Effect of Air Pollution on Self-Reported Mental Health
 - Advised by Dr. Rob Erhardt
- Relevant Coursework: Generalized Linear Models, Bayesian Statistics, Statistical Learning, Time Series, Mathematical Biology, Causal Inference, Stochastic Processes

Davidson College

August 2014 - May 2018

B.A. Economics; Minor in Applied Mathematics, *cum laude*

- Relevant coursework: Microeconomic Theory, Macroeconomic Theory, Environmental Economics, Multivariable Calculus, Linear Algebra, Econometrics, Probability, Mathematical Statistics, Mathematical Modeling, Health Economics, Game Theory, Computational Economics

EMPLOYMENT

Teaching Fellow, Boston University School of Public Health

May 2026 - Present

- Support faculty in preparing education materials and integrating active learning and applied projects to Biostatistics courses and MS Biostatistics capstone
- Co-instructor for BS 800: Accelerated Statistical Training, which is a two week course for MS Applied Biostatistics students that covers probability, statistical inference, and computing in R and SAS
- Prepared slides and gave lectures on Hypothesis testing, Correlation and Regression, and Introduction to R for BU's Summer Institute of Biostatistics program as well as "Reproducibility in Research" and "Anatomy of a Journal Article" in Biostatistics Capstone course

Graduate Research Assistant, Boston University School of Public Health *June 2022 - April 2026*

- Research in spatial statistics and modeling health outcomes
- Lead manuscript development; conduct literature reviews; perform data management and cleaning; statistical analysis tasks

Research Assistant, Wake Forest Baptist Health

February 2021 - May 2022

- Conducted research into how healthcare usage has changed with Medicaid expansion by performing statistical analysis of large healthcare datasets in Stata
- Constructed datasets used to analyze hospitalization data; produced tables and figures to effectively communicate trends present in hospitalization data

Data Analyst, NORC

June 2018 - May 2022

- Served as a contractor for the Economics Unit of the Office of the Investor Advocate at the Securities and Exchange Commission.
- Developed econometric models to analyze investor decision-making and issues in financial markets
- Cleaned and performed statistical analysis on survey data and financial datasets

RESEARCH INTERESTS

- **Methodology**
Causal inference, causal machine learning, Bayesian statistics, Bayesian nonparametrics, spatial statistics, dynamic treatment regimes
- **Applications**
Health policy, population health, spatial and environmental epidemiology, electronic health records

HONORS & AWARDS

- Distinguished Student Paper Award - *ENAR 2026 Spring Meeting*
- Student Travel Award (Fall 2024) - *Boston University School of Public Health*

PUBLICATIONS

1. **Jain A.**, LaValley M., Dukes K., Mohammed S. (2026) Modeling bounded well-being indices using Bayesian double generalized beta regression with spatial and temporal borrowing. *Accepted at Spatial Statistics*.
2. **Jain A.**, Lugo N, Bautista-Hurtado D, Zuniga S, Penugonda A, Jean Philippe NJ, Labossiere S, Pluta KP, Mitchell S, Jack BW, Fernandez A, Drainoni ML, Salloum R, Austad K. (2026) A discrete choice experiment to understand hospital discharge preferences. *Accepted at Journal of General Internal Medicine*.
3. Alperen, J., LaValley, M., Ni, P., Tripp, T., Emig, D., Patil, P., Mohammed, S., **Jain A.**, Rickles M., Thompson E., Davis B., Sullivan S., Acker M., Dukes K. (2026) Characterizing U.S. Community Well-Being Indices: A Methodological Synthesis and Measurement Systems Perspective. *Population Health Metrics*.
4. Dukes, K., Ni, P., Alperen, J., Cesare, N., LaValley, M., Tripp, T., Lane, K., Mohammed, S., Patil, P., Emig, D., Winter, M., Davis B., Wang B., **Jain A.**, Acker M., and Rickles M. (2025) A national growth mixture modeling analysis of county-level COVID-19 incidence rate trajectories and health inequities during three successive pandemic waves in 2020. *Scientific Reports*.
5. **Jain A.**, LaValley M., Dukes K., Lane K., Winter M., Spangler K.R., Cesare N., Wang B., Rickles M., and Mohammed S. (2024) Modeling health and well-being measures using ZIP Code spatial neighborhood patterns. *Scientific Reports*.

WORKING PAPERS

1. **Jain A.**, Jin W., Mohammed S., Lodi S., (2026+) Bayesian causal forests for joint treatments: an analysis of neighborhood food environments on obesity rates with interference. *In preparation*.

2. **Jain A.**, Mohammed S., Lodi S., Jin W., (2026+) Joint BART model with latent factors for causal inference in clustered observational studies. *In preparation.*
3. **Jain A.**, O'Donnell A., Madenci A., Lodi S., (2026+) Practical issues in generating a bootstrap distribution for causal effects of longitudinal treatment strategies. *In preparation.*

CONFERENCE PRESENTATIONS

Conferences (invited, contributed, poster)

1. Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference. Contributed presentation at Joint Statistical Meetings, Boston, MA. *August 2026*
2. Modeling bounded well-being indices using Bayesian double-generalized beta regression with spatial and temporal borrowing. Contributed presentation at 2026 Eastern North American Region Spring Meeting, Indianapolis, USA. *March 2026*
3. Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Contributed presentation at 2025 Eastern North American Region Spring Meeting, New Orleans, USA. *2025 March*
4. Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Contributed presentation at Joint Statistical Meetings, Portland, USA. *August 2024*
5. Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns. Contributed presentation at Joint Statistical Meetings, Toronto, Canada. *August 2023*
6. Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns: a case study on Massachusetts and Georgia. Poster session presentation at New England Statistics Symposium, Boston, MA. *June 2023*
7. Regression-Based Oversampling for Mutual Fund Owners. Contributed poster presentation at Joint Statistics Meetings, Virtual. *August 2020*

General audience

1. Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference. Poster presentation at Boston University Population Health Data Science Poster Session, Boston, MA. *September 2026*
2. Performance of “bag of little bootstrap” in simulated complex longitudinal data. Kolokotronis Circle presentation at CAUSALab, Harvard T.H. Chan School of Public Health, Boston, MA. *September 2025*
3. Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Poster presentation at Boston University Population Health Data Science Poster Session, Boston, MA. *September 2025*
4. Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns. Poster presentation at Boston University Population Health Data Science Poster Session, Boston, MA. *September 2023*

TEACHING EXPERIENCE

Teaching Fellow

Department of Biostatistics, Boston University School of Public Health

- SPH BS 880 — Biostatistics Capstone: Design and Analysis of Investigations *Fall 2026*

- SPH BS 755 — Theory of Linear Models in Biostatistics *Fall 2026*
- SPH BS 800 — Accelerated Statistical Training *Summer 2026*
- SPH BS 401S — Survey in Biostatistical Methods *Summer 2026*

Teaching Assistant

Department of Biostatistics, Boston University School of Public Health

- SPH BS 849 — Bayesian Modeling for Biomedical Research & Public Health *Spring 2026*
- SPH BS 807 — Applied Causal Inference in Health Research *Fall 2024*

Department of Mathematics and Statistics, Wake Forest University

- STA 111 — Elementary Probability and Statistics *Fall 2020*

Department of Economics, Davidson College

- ECO 105 — Statistics and Basic Econometrics *Spring 2018*

Tutor

Department of Mathematics and Statistics, Wake Forest University

- Math & Stats Center *Fall 2020*

SERVICE

Session Chair - Bayesian Methods for Categorical and Survey Data at JSM *August 2026*

Officer - Boston University Student Chapter of the ASA *March 2024-Present*

Student representative - Biostatistics Graduate Education Committee *June 2024-Present*

Volunteer academic mentor - Boston Partners in Education *February 2024-December 2025*

Lead student volunteer - New England Statistics Symposium *May 2023*

SKILLS

Programming Languages and Software

R, Stata, Python, SAS, SQL, GitHub, Excel, Word, PowerPoint, Visio, Final Cut Pro

Languages

Hindi (speaking)